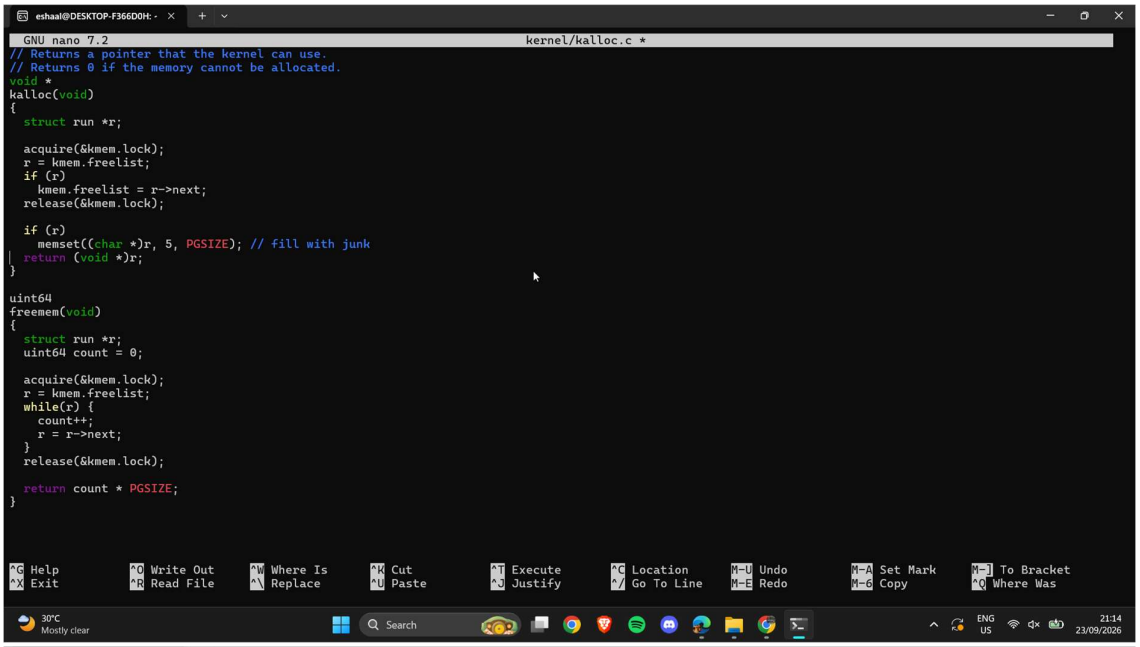
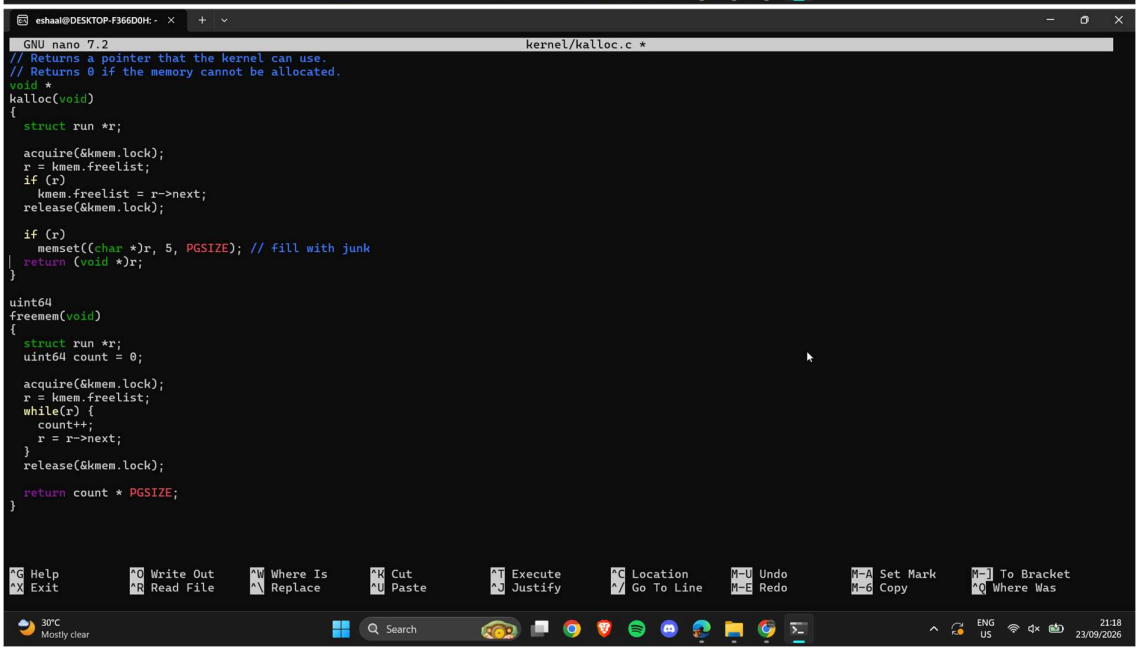
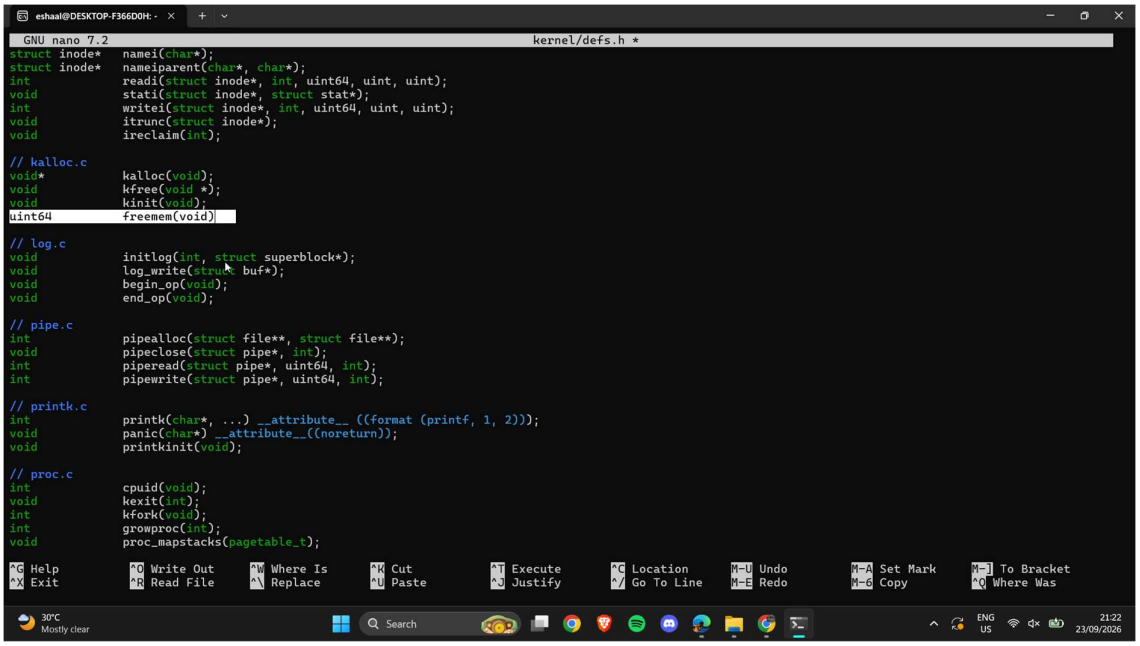
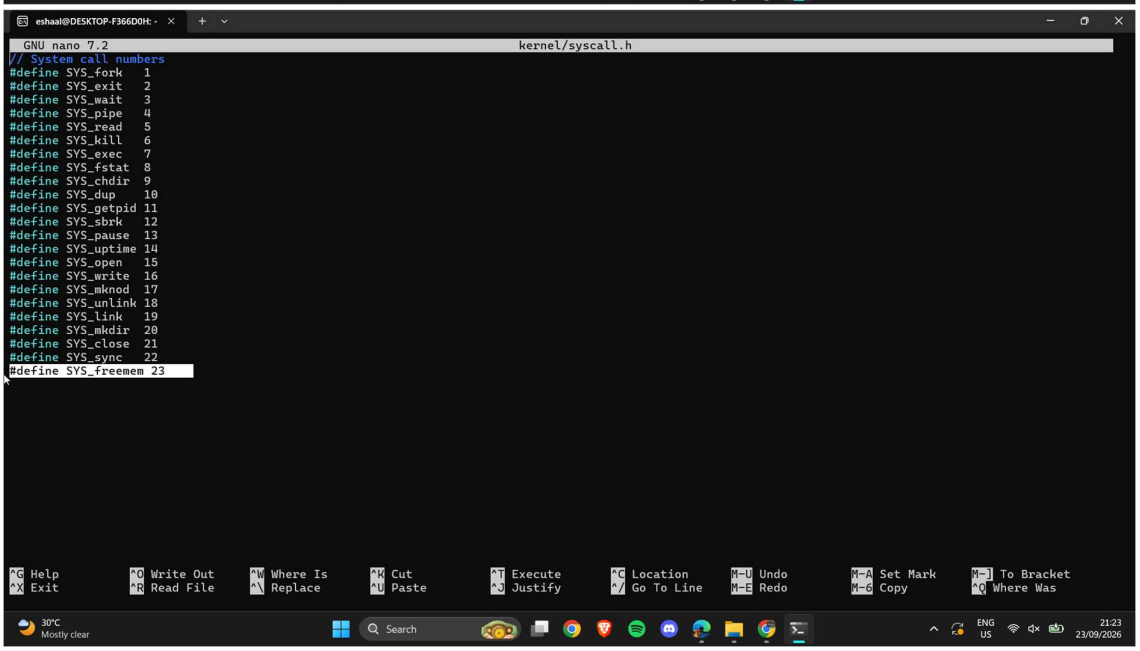
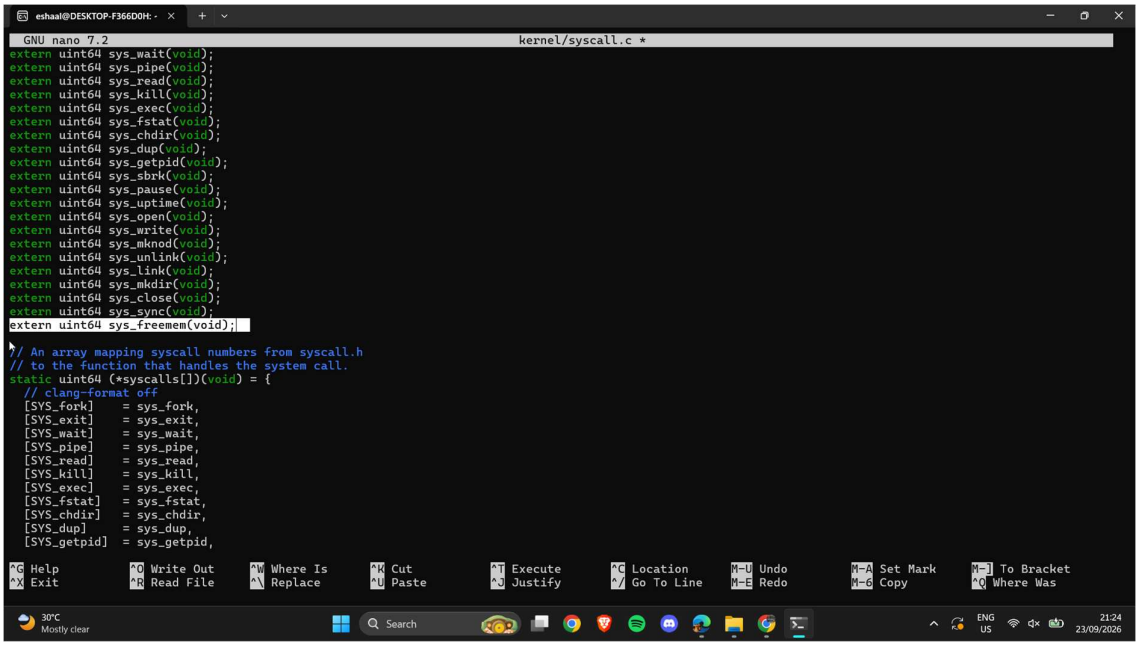
S.Eshaal Gardezi (30868)

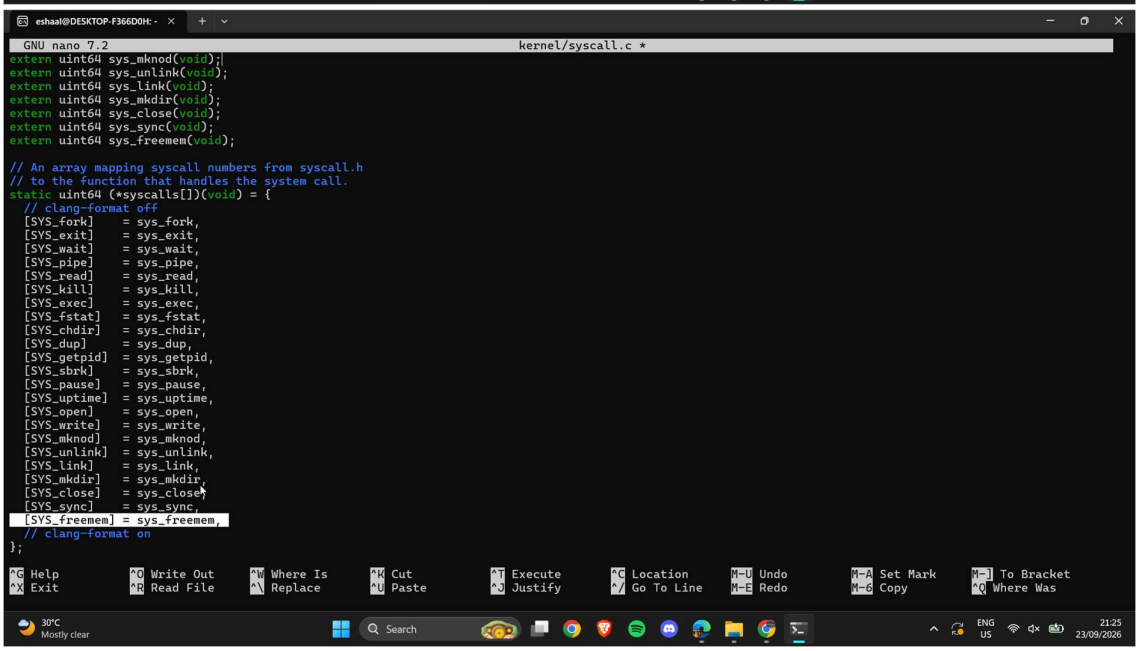





eshaal@DESKTOP-F366D0H: ~
GNU nano 7.2 kernel/defs.h *
struct inode* namei(char*);
struct inode* nameiparent(char*, char*);
int readi(struct inode*, int, uint64, uint, uint);
void stati(struct inode*, struct stat*);
int writei(struct inode*, int, uint64, uint, uint);
void itrunc(struct inode*);
void ireclaim(int);
// kalloc.c
void* kalloc(void);
void kfree(void *);
void kinit(void);
uint64 freemem(void)
// log.c
void initlog(int, struct superblock*);
void log_write(struct buf*);
void begin_op(void);
void end_op(void);
// pipe.c
int pipealloc(struct file**, struct file**);
void pipeclose(struct pipe*, int);
int piperead(struct pipe*, uint64, int);
int pipewrite(struct pipe*, uint64, int);
// printk.c
int printk(char*, ...) __attribute__ ((format (printf, 1, 2)));
void panic(char*) __attribute__((noreturn));
void printkinit(void);
// proc.c
int cpuid(void);
void kexit(int);
int kfork(void);
int growproc(int);
void proc_mapstacks(pagetable_t);
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was
30°C Mostly clear
Search
ENG US
21:22 23/09/2026


eshaal@DESKTOP-F366D0H: ~
GNU nano 7.2 kernel/syscall.h
// System call numbers
#define SYS_fork 1
#define SYS_exit 2
#define SYS_wait 3
#define SYS_pipe 4
#define SYS_read 5
#define SYS_kill 6
#define SYS_exec 7
#define SYS_fstat 8
#define SYS_chdir 9
#define SYS_dup 10
#define SYS_getpid 11
#define SYS_sbrk 12
#define SYS_pause 13
#define SYS_uptime 14
#define SYS_open 15
#define SYS_write 16
#define SYS_mknod 17
#define SYS_unlink 18
#define SYS_link 19
#define SYS_mkdir 20
#define SYS_close 21
#define SYS_sync 22
#define SYS_freemem 23
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was
30°C Mostly clear
Search
ENG US
21:23 23/09/2026

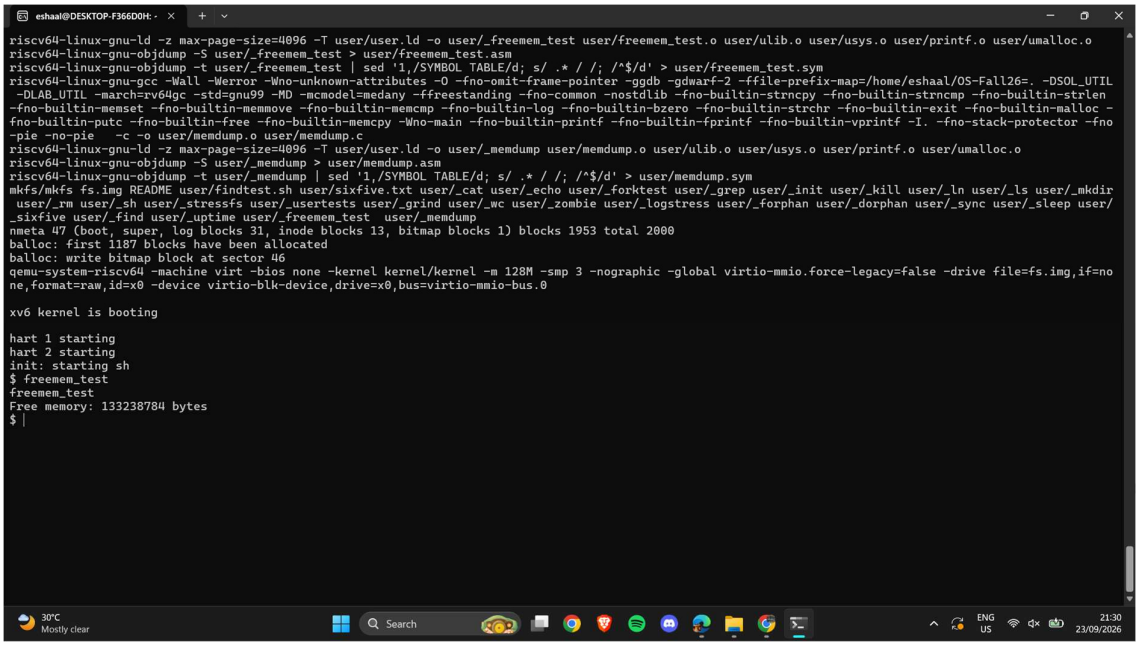

eshaal@DESKTOP-F366D0H: ~
GNU nano 7.2 kernel/syscall.c *
extern uint64 sys_wait(void);
extern uint64 sys_pipe(void);
extern uint64 sys_read(void);
extern uint64 sys_kill(void);
extern uint64 sys_exec(void);
extern uint64 sys_fstat(void);
extern uint64 sys_chdir(void);
extern uint64 sys_dup(void);
extern uint64 sys_getpid(void);
extern uint64 sys_sbrk(void);
extern uint64 sys_pause(void);
extern uint64 sys_uptime(void);
extern uint64 sys_open(void);
extern uint64 sys_write(void);
extern uint64 sys_mknod(void);
extern uint64 sys_unlink(void);
extern uint64 sys_link(void);
extern uint64 sys_mkdir(void);
extern uint64 sys_close(void);
extern uint64 sys_sync(void);
extern uint64 sys_freemem(void);
// An array mapping syscall numbers from syscall.h
// to the function that handles the system call.
static uint64 (*syscalls[])(void) = {
// clang-format off
[SYS_fork] = sys_fork,
[SYS_exit] = sys_exit,
[SYS_wait] = sys_wait,
[SYS_pipe] = sys_pipe,
[SYS_read] = sys_read,
[SYS_kill] = sys_kill,
[SYS_exec] = sys_exec,
[SYS_fstat] = sys_fstat,
[SYS_chdir] = sys_chdir,
[SYS_dup] = sys_dup,
[SYS_getpid] = sys_getpid,
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was
30°C Mostly clear
Search
ENG US
21:24 23/09/2026

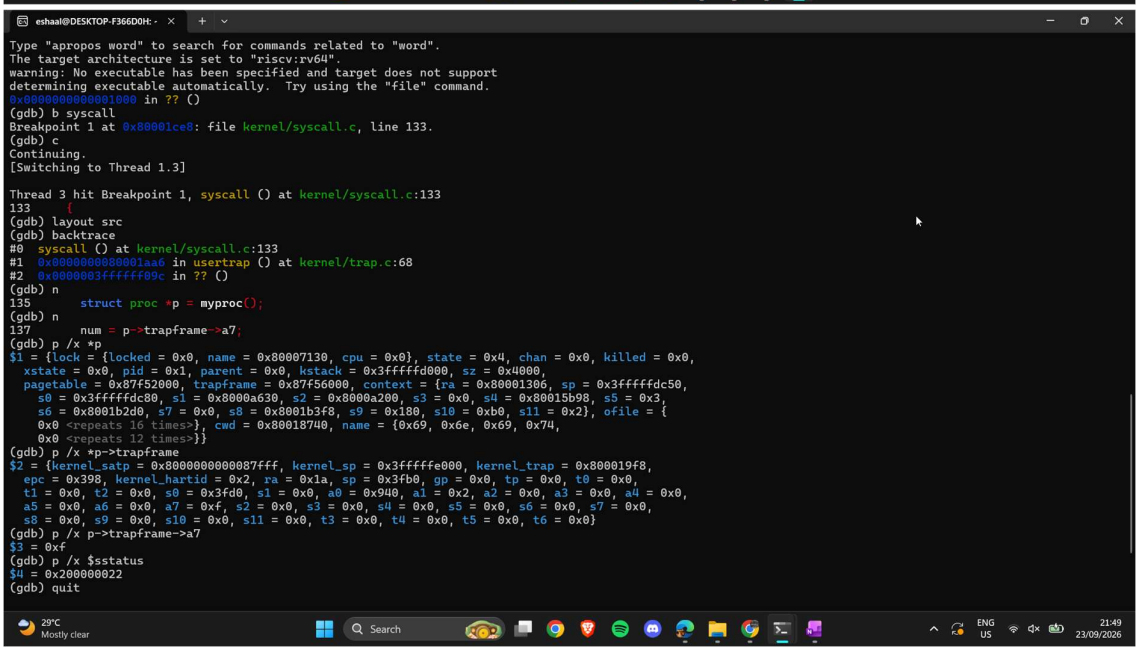

eshaal@DESKTOP-F366D0H: ~
GNU nano 7.2 kernel/syscall.c *
extern uint64 sys_mknod(void);
extern uint64 sys_unlink(void);
extern uint64 sys_link(void);
extern uint64 sys_mkdir(void);
extern uint64 sys_close(void);
extern uint64 sys_sync(void);
extern uint64 sys_freemem(void);
// An array mapping syscall numbers from syscall.h
// to the function that handles the system call.
static uint64 (*syscalls[])(void) = {
// clang-format off
[SYS_fork] = sys_fork,
[SYS_exit] = sys_exit,
[SYS_wait] = sys_wait,
[SYS_pipe] = sys_pipe,
[SYS_read] = sys_read,
[SYS_kill] = sys_kill,
[SYS_exec] = sys_exec,
[SYS_fstat] = sys_fstat,
[SYS_chdir] = sys_chdir,
[SYS_dup] = sys_dup,
[SYS_getpid] = sys_getpid,
[SYS_sbrk] = sys_sbrk,
[SYS_pause] = sys_pause,
[SYS_uptime] = sys_uptime,
[SYS_open] = sys_open,
[SYS_write] = sys_write,
[SYS_mknod] = sys_mknod,
[SYS_unlink] = sys_unlink,
[SYS_link] = sys_link,
[SYS_mkdir] = sys_mkdir,
[SYS_close] = sys_close,
[SYS_sync] = sys_sync,
[SYS_freemem] = sys_freemem,
// clang-format on
};
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was
30°C Mostly clear
Search
ENG US
21:25 23/09/2026

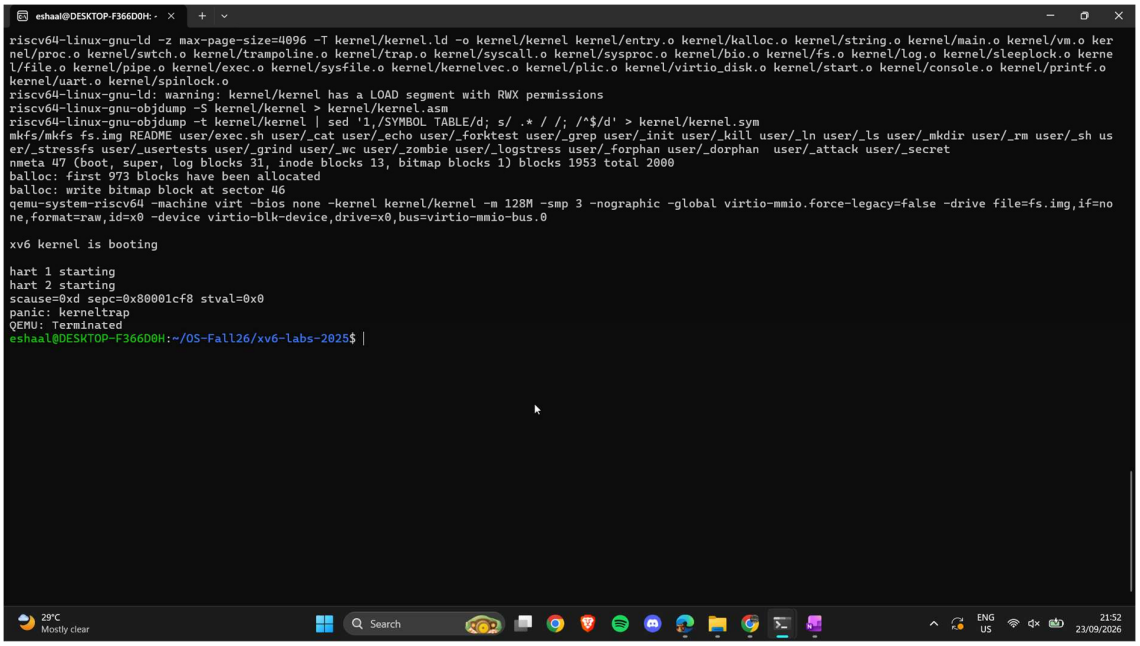

eshaal@DESKTOP-F366D0H: ~
GNU nano 7.2 kernel/sysproc.c *
#include "types.h"
#include "riscv.h"
#include "defs.h"
#include "param.h"
#include "memlayout.h"
#include "spinlock.h"
#include "proc.h"
#include "vm.h"
uint64
sys_freemem(void)
{
return freemem();
}
uint64
sys_exit(void)
{
int n;
argint(0, &n);
kexit(n);
return 0; // not reached
}
uint64
sys_getpid(void)
{
return myproc()->pid;
}
uint64
sys_fork(void)
{
return kfork();
}
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify
ENG US
21:26
23/09/2026

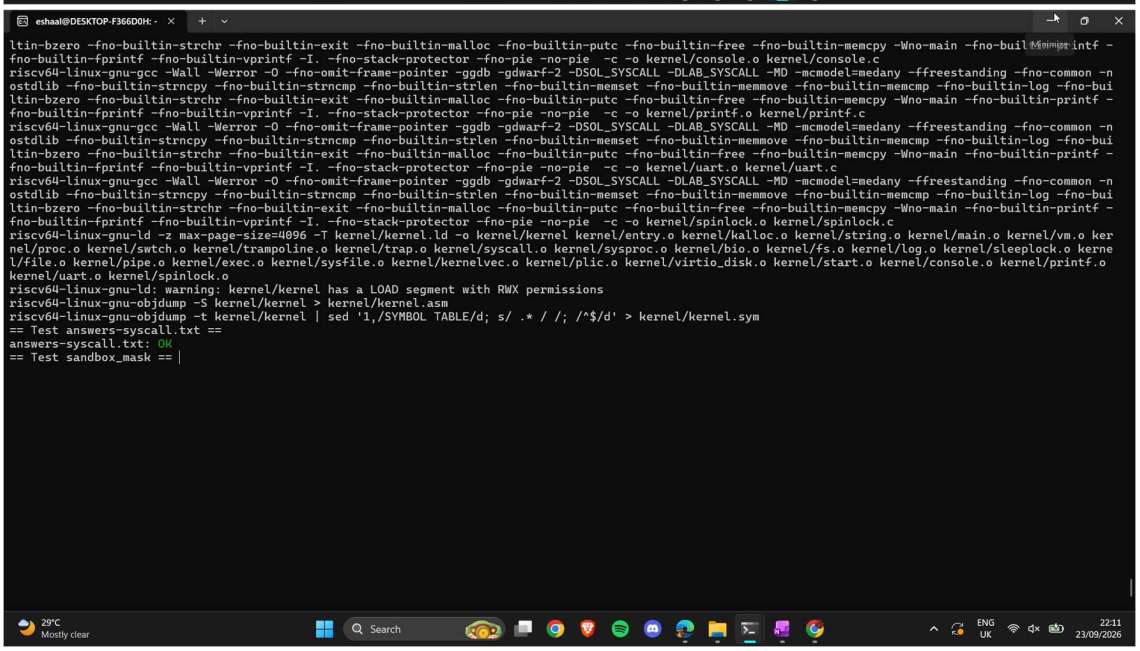

eshaal@DESKTOP-F366D0H: ~
GNU nano 7.2 user/user.h *
#define SBRK_ERROR ((char *)-1)
struct stat;
// system calls
int fork(void);
int exit(int) __attribute__((noreturn));
int wait(int *);
int pipe(int *);
int write(int, const void *, int);
int read(int, void *, int);
int close(int);
int kill(int);
int exec(const char *, char **);
int open(const char *, int);
int mknod(const char *, short, short);
int unlink(const char *);
int fstat(int fd, struct stat *);
int link(const char *, const char *);
int mkdir(const char *);
int chdir(const char *);
int dup(int);
int getpid(void);
char *sys_sbrk(int, int);
int pause(int);
int uptime(void);
int sync(void);
uint64 freemem(void);
// ulib.c
int stat(const char *, struct stat *);
char *strcpy(char *, const char *);
void *memmove(void *, const void *, int);
char *strchr(const char *, char c);
int strcmp(const char *, const char *);
char *gets(char *, int max);
uint strlen(const char *);
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify
ENG US
21:27
23/09/2026

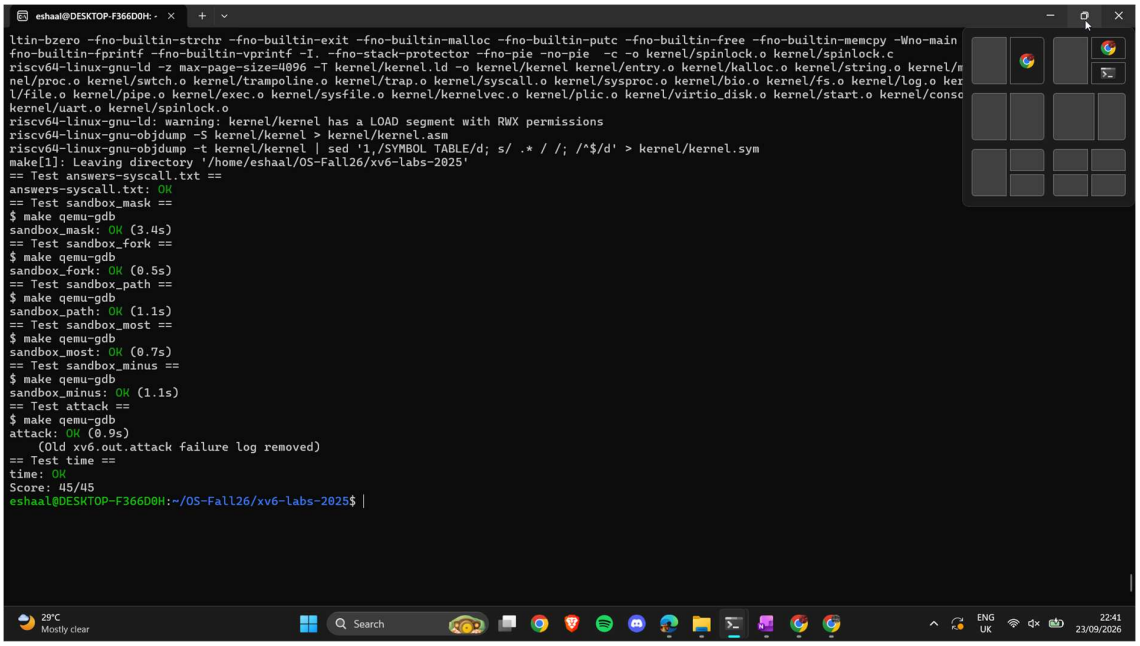

eshaal@DESKTOP-F366D0H: ~
GNU nano 7.2 user/usys.pl *
my $prefix = "sys_";
my $name = shift;
if ($name eq "sbrk") {
print ".global $prefix$name\n";
print "$prefix$name:\n";
} else {
print ".global $name\n";
print "$name:\n";
}
print " li a7, SYS_${name}\n";
print " ecall\n";
print " ret\n";
}
entry("fork");
entry("exit");
entry("wait");
entry("pipe");
entry("read");
entry("write");
entry("close");
entry("kill");
entry("exec");
entry("open");
entry("mknod");
entry("unlink");
entry("fstat");
entry("link");
entry("mkdir");
entry("chdir");
entry("dup");
entry("getpid");
entry("sbrk");
entry("pause");
entry("uptime");
entry("sync");
entry("freemem");
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify
ENG US
21:28
23/09/2026


eshaal@DESKTOP-F366D0H: ~
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_freemem_test user/freemem_test.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_freemem_test > user/freemem_test.asm
riscv64-linux-gnu-objdump -t user/_freemem_test | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/freemem_test.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/memdump.o user/memdump.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_memdump user/memdump.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_memdump > user/memdump.asm
riscv64-linux-gnu-objdump -t user/_memdump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/memdump.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_uptime user/_freemem_test user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1187 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ freemem_test
freemem_test
Free memory: 133238784 bytes
$
30°C
Mostly clear
Search
ENG
US
21:30
23/09/2026


eshaal@DESKTOP-F366D0H: ~
Type "apropos word" to search for commands related to "word".
The target architecture is set to "riscv:rv64".
warning: No executable has been specified and target does not support
determining executable automatically. Try using the "file" command.
0x0000000000001000 in ?? ()
(gdb) b syscall
Breakpoint 1 at 0x80001ce8: file kernel/syscall.c, line 133.
(gdb) c
Continuing.
[Switching to Thread 1.3]
Thread 3 hit Breakpoint 1, syscall () at kernel/syscall.c:133
133 {
(gdb) layout src
(gdb) backtrace
#0 syscall () at kernel/syscall.c:133
#1 0x0000000080001aa6 in usertrap () at kernel/trap.c:68
#2 0x0000003fffffff09c in ?? ()
(gdb) n
135 struct proc *p = myproc();
(gdb) n
137 num = p->trapframe->a7;
(gdb) p /x *p
$1 = {lock = {locked = 0x0, name = 0x80007130, cpu = 0x0}, state = 0x4, chan = 0x0, killed = 0x0,
xstate = 0x0, pid = 0x1, parent = 0x0, kstack = 0x3fffffd000, sz = 0x4000,
pagetable = 0x87f52000, trapframe = 0x87f56000, context = {ra = 0x80001306, sp = 0x3fffffdc50,
s0 = 0x3fffffdc80, s1 = 0x8000a630, s2 = 0x8000a200, s3 = 0x0, s4 = 0x80015b98, s5 = 0x3,
s6 = 0x8001b2d0, s7 = 0x0, s8 = 0x8001b3f8, s9 = 0x180, s10 = 0xb0, s11 = 0x2}, ofile = {
0x0 <repeats 16 times>}, cwd = 0x80018740, name = {0x69, 0x6e, 0x69, 0x74,
0x0 <repeats 12 times>}}
(gdb) p /x *p->trapframe
$2 = {kernel_satp = 0x8000000000087fff, kernel_sp = 0x3fffffe000, kernel_trap = 0x800019f8,
epc = 0x398, kernel_hartid = 0x2, ra = 0x1a, sp = 0x3fb0, gp = 0x0, tp = 0x0, t0 = 0x0,
t1 = 0x0, t2 = 0x0, s0 = 0x3fd0, s1 = 0x0, a0 = 0x940, a1 = 0x2, a2 = 0x0, a3 = 0x0, a4 = 0x0,
a5 = 0x0, a6 = 0x0, a7 = 0xf, s2 = 0x0, s3 = 0x0, s4 = 0x0, s5 = 0x0, s6 = 0x0, s7 = 0x0,
s8 = 0x0, s9 = 0x0, s10 = 0x0, s11 = 0x0, t3 = 0x0, t4 = 0x0, t5 = 0x0, t6 = 0x0}
(gdb) p /x p->trapframe->a7
$3 = 0xf
(gdb) p /x $sstatus
$4 = 0x200000022
(gdb) quit
29°C
Mostly clear
Search
ENG
US
21:49
23/09/2026


eshaal@DESKTOP-F366D0H: ~
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 973 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 1 starting
hart 2 starting
scause=0xd sepc=0x80001cf8 stval=0x0
panic: kerneltrap
QEMU: Terminated
eshaal@DESKTOP-F366D0H:~/OS-Fall26/xv6-labs-2025$
29°C
Mostly clear
Search
ENG
US
21:52
23/09/2026


eshaal@DESKTOP-F366D0H: ~
ltin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/console.o kernel/console.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/printf.o kernel/printf.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/uart.o kernel/uart.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
29°C
Mostly clear
Search
ENG
UK
22:11
23/09/2026


eshaal@DESKTOP-F366D0H:
ltin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main
fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/m
nel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o ker
l/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/consc
kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/eshaal/OS-Fall26/xv6-labs-2025'
== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (3.4s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (0.5s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (1.1s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (0.7s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.1s)
== Test attack ==
$ make qemu-gdb
attack: OK (0.9s)
(Old xv6.out.attack failure log removed)
== Test time ==
time: OK
Score: 45/45
eshaal@DESKTOP-F366D0H:~/OS-Fall26/xv6-labs-2025$
29°C
Mostly clear
Search
ENG
UK
22:41
23/09/2026